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ANTI-CORRUPTION
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ARTIFICIAL INTELLIGENCE IN ANTI-MONEY LAUNDERING

STRENGTHENING THE FIGHT AGAINST CORRUPTION AND ILLICIT FINANCIAL FLOWS

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Abstract

Money laundering is one of the key enablers of corruption, undermining the rule of law, economic stability, and the integrity of the global financial system. Despite the existing regulatory and institutional framework in combating money laundering, the effectiveness of the current anti-money laundering (AML) regime is continuously questioned, especially due to high compliance costs, fragmented regulatory approaches, overreliance on manual and rule-based systems, and limited measurable benefits of AML for crime reduction. This paper focuses on the structural weaknesses of the existing AML framework and explores whether artificial intelligence (AI) can contribute to a more effective and proportionate response. Building on a cost-benefit analysis of AML compliance and an examination of the root causes of systemic inefficiencies, the paper assesses the potential of AI technologies, such as machine learning, natural language processing, and generative and agentic AI, to enhance customer due diligence, transaction monitoring, risk assessment, and international cooperation. Through selected case studies, including vendor-based AML solutions and international public-sector initiatives, the paper illustrates how AI-driven tools can reduce false positives, improve analytical capacity, and allow human expertise to focus on higher-value investigative tasks. At the same time, the paper critically examines the risks associated with AI deployment in AML, including bias, lack of explainability, and data quality concerns. It concludes that AI is not the ultimate remedy, but if deployed responsibly, with strong safeguards, transparency, and human oversight - as a crucial part of AI deployment in AML, it can significantly strengthen the AML regime and contribute to a more effective prevention of money laundering.

Key words: Money laundering, Anti-Money Laundering, Corruption, Artificial Intelligence, Effectiveness

About the Author

Alida Radončić is a legal professional from Montenegro with deep expertise in legal frameworks with a focus on anticorruption, organized crime, rule of law, and EU-aligned standards. Her experience spans drafting legislation, conducting complex legal analysis, and supporting projects within judicial and law enforcement institutions. One of her specific interests includes anti-money laundering compliance, especially in the context of emerging technologies.

Summary

This research paper examines the effectiveness of the current anti-money laundering (AML) regime and explores the potential of artificial intelligence (AI) to strengthen efforts against illicit financial flows, in particular, money laundering. Basing the research paper on desk review methodology, the study analyses existing academic literature, policy documents, international standards, regulatory reports, and selected case studies to assess structural weaknesses in the AML regime and emerging technological responses.

The analysis identifies persistent shortcomings in the current AML framework, including high compliance costs, fragmented regulatory approaches, overreliance on manual and rule-based systems, and limited measurable impact on reducing money laundering activities. While AML compliance is widely recognized as essential for safeguarding the financial system and the rule of law, cost-benefit assessments indicate that the effectiveness of existing measures remains uncertain. Factors such as legacy systems, de-risking practices, lack of regulatory harmonization, and inadequate monitoring contribute to systemic inefficiencies.

The paper highlights the growing relevance of AI as a tool to address these challenges. AI-based solutions demonstrate significant potential to enhance customer due diligence, transaction monitoring, risk assessment, and international cooperation. Desk-reviewed case studies from both private-sector vendors and international initiatives illustrate that AI can reduce false positives, improve detection of complex laundering schemes, and enable human analysts to focus on higher-value analytical tasks.

At the same time, the research underscores risks associated with AI deployment in AML, including data bias, lack of transparency, explainability challenges, and governance concerns. The study concludes that AI is not a substitute for human judgment but, when implemented responsibly with robust safeguards and human oversight, can substantially improve the effectiveness and resilience of the AML system in combating corruption and illicit financial flows.

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1. Introduction

Money laundering remains a central enabler of corruption, facilitating the concealment of illicit funds derived from various predicate offences. Taking into consideration the so-called “chameleonic” threat that money laundering poses (Bakardjieva Engelbrekt *et al.*, 2024), and the evolving social norms (such as technological advancements) which facilitate these illegal practices, one must ask: *quo vadis* AML system?

This question of the effectiveness of the anti-money laundering measures is repeatedly asked, specifically, how dissuasive these measures really are in combating serious crime such as money laundering (Ferwerda, 2018).

However, with the ever-growing usage of artificial intelligence tools in all areas of life, but also for criminal purposes, this research paper aims to explore the possibilities of leveraging AI to enhance the anti-money laundering response.

Traditional anti-money laundering (AML) systems, heavily reliant on manual reporting, are increasingly unable to detect complex and rapidly evolving laundering schemes. Therefore, taking into account the omnipresence of money laundering, as well as high compliance costs and a lack of harmonized approach in implementing anti-money laundering measures (especially in the EU), there is a growing need to address the threats that money laundering poses more effectively. Still, AI deployment carries inherent risks, which ought to be mitigated with strong ethical basis of the deployed AI tool, and rooted in robust regulatory guidelines, with human oversight as a crucial component.

2. The nature of money laundering

2.1. Elements of the criminal offence of money laundering

Money laundering is a well-known concept to everyday clients in banks, to compliance officers, and financial investigators. Due to the omnipresence of money laundering as a phenomenon, but also regulatory divergences, there are numerous definitions. Starting with fairly simple ones, such as 'turning dirty money to clean money' (Amrani, 2012), to more technical definitions set by international regulatory frameworks. Still, money laundering can be narrowed down to a process by which illegal proceeds (derived from criminal activity) are disguised to conceal their illicit origin and reintroduced into legitimate capital flows.

Due to its many forms, from a legal point of view, there might be some overlap of several acts that constitute the *actus reus* of the criminal offence of money laundering. While taking into account definitions of money laundering, either the one from the European Convention (1990) or the United Nations Convention (1988), the commonality they share is that the act of the criminal offence of money laundering consists of: conversion or transfer of illicit property; concealment or disguise; acquisition, possession, or use; participation or facilitation. On the other hand, all of these constitutive acts of money laundering can be divided into the so-called three-stage model, which usually includes: placement, layering, and integration.

In addition to this, there are numerous techniques of money laundering. For instance, smurfing or structuring is the most common one, since it consists of depositing smaller amounts of money so that compliance thresholds are avoided. Another known technique is the use of shell companies with fictitious business and invoices, or the buying of luxury goods in order to conceal the illegitimate source of money, and many others (Fenergo, 2025).

Precisely because of the multi-faceted nature of money laundering, which makes its countering all the more difficult, which justifies the etiquette of the so-called chameleonic nature of money laundering due to the possibility of adapting to changing circumstances, it is clear that combating money laundering is burdensome.

Therefore, even though it is noted as a seemingly simple concept (Schott, 2006), it causes not-so-simple ripple effects with real-world consequences on society's economy and the rule of law. If we take a functioning rule of law system and a thriving economy of a country as the cornerstones of a welfare country, then money laundering, as one of the profound destabilizing factors affecting these two pillars, is all the more concerning. In addition, money laundering is placed as the most common form of illicit financial flows (Palan, Chavagneux and Murphy, 2013), with the official statistics showcasing that criminal proceeds laundered each year amount to between 2 % and 5 % of global GDP.

The fact that the AML system still has a long way to go in preventing the illegitimate money flow is showcased by additional efforts made by countries and international organisations to combat money laundering, mostly because of the recognized threat that it poses to society and the global financial system. These efforts have resulted in implementing numerous legal and regulatory measures that are directed at preventing the proceeds of crime from entering the financial system (Nazzari and Reuter, 2025).

With this said, the handling of this phenomenon seems very much like a Rubik's Cube, with numerous possibilities that could result in a solution. However, is there only one solution to the puzzle that is anti-money laundering, or could several combinations work? To answer this, the current combination that is in use - the AML compliance regime needs deconstructing.

3. Effectiveness of the current AML system

3.1. Cost-benefit assessment

The matter of the effectiveness of the anti-money laundering system is not a novel issue. Even back when the first anti-money laundering system was established due to the emergence of illegal proceeds from drug trafficking, there were perspectives of possible consequences on doing everyday business due to cumbersome compliance obligations.

One premise that this research paper rests upon is the ineffectiveness of the current AML system. However, this (in)effectiveness is not due to the lack of

proper technology but to a variety of reasons, while proper technology could help mitigate some of these causes. If these reasons were to be put on a bulleted list, it would be non-exhaustive, as many other factors can affect the effectiveness of the AML system - even the political situation in a country (UNODC, 2009). This paper will focus on some of the possible factors affecting the AML effectiveness, or even more so, the root causes of its ineffectiveness.

Foremost, to answer the question of the effectiveness of the anti-money laundering measures and how dissuasive these measures really are, a cost-benefit perspective is useful for this purpose. For instance, in a study on the effectiveness of anti-money laundering policy, a cost-benefit perspective for AML compliance was pretty indicative of the state of affairs in AML compliance. In this study, it was demonstrated that cost-benefit analyses are common in assessing proposed policies in all fields, but for anti-money laundering, this is rare. This is an indication of how difficult it is to actually assess the true costs versus benefits of AML compliance. In summary, the study presented the costs of policy-making in anti-money laundering for each obliged entity, including law enforcement and the judiciary. However, when it comes to benefits, the study concludes that these remain unclear and underlines a message stated by another cited author, Whitehouse, who said that it would take a brave person to say that the price of countering serious crime is too high (Ferwerda, 2018).

For instance, in the recently published study on the effectiveness of the current AML system, while the authors stress that their study doesn't provide a comprehensive assessment of how well the system works, they state that their review of the available data supports the view that AML controls may have a limited impact on crime levels (Nazzari and Reuter, 2025). However, an impression gained from the conclusions of these studies is that, regardless of the costs it takes to tackle money laundering, the benefits need not be clear, but they are not as important as trying to combat crime such as money laundering and minimize its negative societal impact. This also means that there is a space for changing *how* money laundering is prevented.

3.2. Root causes of the AML ineffectiveness

Still, it is a well-known phenomenon that the AML system puts a lot of burden on both the obliged entities as well as third parties (legitimate customers) regarding the compliance costs. AML compliance costs are the key driving force behind the scrutiny of the (in)effectiveness of the current AML system. From compliance costs for obliged entities, which lead to cumbersome administrative burdens, to disruptions in doing business in the financial sector, mostly due to AML measures such as the customer identification (Palazov, 2021). Besides the costs for the obliged entities, the costs for honest customers, *i.e.*, legitimate users of the financial services, are also apparent. For instance, due to the know your customer (KYC) processes, legitimate financial transactions can be delayed, which could have serious consequences on legitimate clients and their business in terms of delaying an important business deal. (Ferwerda, 2018).

The possibility of ineffectiveness of the AML system was pointed out back in the nineties by relevant experts, who commented on the EU's Directive on the prevention of money laundering as being too costly due to numerous regulatory obligations, which could dissuade clients from doing business with entities that have strict regulatory compliance. (Alford, 1994). That these predictions are not a relic of the past is evident from studies that show regulatory compliance costs are consistently high, with a study from 2023 revealing that financial crime compliance costs increased for 98% of financial institutions in 2023 and that the total cost of financial crime compliance in the EMEA region has reached \$85 billion (LexisNexis, 2023). Furthermore, capital markets might disrupt AML implementation due to globalization and liberalization of capital movements, which could cause less effective AML compliance because of the divergences in national regulatory frameworks.

In addition to this, overreliance on manual reporting systems, or the so-called legacy systems, leads to a less dynamic analysis, unable to adapt to new risks or emerging circumstances that affect compliance processes. As defined by the Financial Action Task Force (FATF), legacy systems are "low-tech" systems, meaning that most of the work regarding data entry is still done via human input (FATF, 2021).

Also, as entailed by FATF, AML compliance often applies one-size-fits-all controls, which pushes institutions to de-risk by cutting off high-risk customers rather than managing risk. As pointed out by numerous authors who reflected on the consequences of de-risking (Palazov, 2021), it is clear that such measures can force legitimate clients into less regulated channels and undermine financial inclusion. These practices increase the risk of the emergence of unique forms of money laundering, such as the “bank capture”, which consists of having control of financial entities in unregulated or weakly regulated jurisdictions so that funds can move without effective AML oversight (Anti-Money Laundering Compliance Unit, 2021).

In addition, regulatory lapses that have occurred in practice, such as the Danske Bank case, showcase how the current AML system still faces insufficient mitigating measures. The Danske Bank concerns abuse of the banking system for the purpose of money laundering, for which the Estonian branch of the Danske Bank was utilized to transfer over 800 billion euros in suspicious transactions originating from Estonia, Russia, and Latvia. Despite the warnings by the regulatory bodies and internal auditors, Danske Bank didn't manage to address numerous mishaps in risk management, new client identification, and monitoring of transactions. (Elucidate, 2023) Besides this infamous case that had real-world implications on how AML compliance is perceived, even recent cases point to the same deficiencies in AML systems that exist to this day. For instance, more recent example includes the case of Deutsche Bank and its US affiliates, in which they were fined \$186 million because AML issues highlighted in earlier U.S. central bank orders were not fixed (Reuters, 2023).

In addition, as the international standard-setter in the field of anti-money laundering, the FATF evaluations consistently show that investigating, prosecuting, and sanctioning money laundering remains one of the weakest areas worldwide (FATF *et al.*, 2025). Finally, due to the increased globalisation, criminals move their money more effortlessly, utilizing both the technological advancements as well as the different regulatory frameworks. All of these factors contribute to the ineffectiveness of the current AML system.

Still, regardless of the high-compliance costs and administrative burdens that AML poses for financial compliance professionals, a LexisNexis study conducted in 2017 presented results that indicate that compliance is viewed as a strategic asset for businesses to thrive and protect their integrity in the market, and not just a requirement. (LexisNexis Study, 2017) Therefore, there is a space for introducing new approaches to AML compliance to financial regulators, with the use of AI tools being the natural win-win, meaning that both compliance costs could be reduced and businesses could continue to grow and be protected from market disruptors, such as the illegitimate financial flows.

4. Quo vadis, AML?

4.1. AML regime in the light of technological developments

It is clear that due to the many faces of money laundering, its tracing is harder, especially with the ever-growing use of technological tools. Specifically, in the recently issued opinion of the European Banking Authority, it is stated that criminals exploit AI to automate money laundering, hide fund origins, and evade detection through fake documents and deepfakes. Financial institutions struggle to detect these fast, complex schemes, requiring advanced technology and expert skills to counter them. In addition, EBA noted that careless RegTech deployment has heightened exposure to money laundering (EBA, 2025). Therefore, the best way to tackle this issue, but also to increase the effectiveness of the existing AML system, is to "fight" back with responsible deployment of AI tools.

The FATF identified challenges related to the development, adoption and application of these innovative solutions or practices. Many of these challenges are due to outstanding operational and regulatory constraints, such as legacy AML/CFT compliance and traditional regulatory frameworks and oversight mechanisms. The complexities and costs involved in replacing or updating legacy systems make it challenging to exploit the potential of innovative approaches to AML/CFT for both industry and government. For the industry, the cost-benefit analysis to adopt new technologies continues to be an obstacle to greater uptake of innovative solutions for AML/CFT, based in

part on a real or perceived lack of regulatory incentives to pursue innovation. Difficulties with the explainability and interpretability of digital solutions are another key challenge for both industry and regulators that in part stems from the limited availability of relevant expertise and a lack of awareness of innovative technologies' potential among AML/CFT professionals, both in industry and government (FATF 2021).

Nevertheless, support for adopting new technologies in addressing money laundering from relevant international addresses is not scarce. For instance, the FATF endorsed responsible innovation back in 2017, expressing strong support for exploring new opportunities in technological development that could improve the effectiveness of AML/CFT measures. Also, the UN with its Security Council Resolution 2462(2019) (UN, 2019[2]) called upon all States to exploit the use of new and emerging financial and regulatory technologies to bolster financial inclusion.

5. AI and AML

5.1. Potential of AI in AML

Criminal networks are increasingly using advanced technologies to bribe, launder, collude, and conceal, often faster and more effectively than the systems meant to stop them. In this context, the use of tools like artificial intelligence (AI) and big data by public authorities, companies, and civil society is not a luxury but a necessity (OECD 2025). That money laundering, especially the transnational forms, needs tackling by emerging technologies such as AI, is therefore evident from the rapid digitalisation of services, which resulted in a greater sophistication of the *modus operandi* of economic crimes. In addition to this, there is an increase in international requests, which also requires investing in tools that can aid in speeding up responses to these requests (FATF 2025).

While artificial intelligence (AI) is widely recognized in today's technology landscape, its potential application within the field of anti-money laundering (AML) compliance is not entirely new. Rather, the AML sector has long been in search of solutions to address the limitations of traditional compliance frameworks. For instance, a 2016 study recognized the need to replace

traditional, manual AML systems, and due to the multi-faceted nature of money laundering, which requires automated approaches in its detection and prevention. The study also recognized the lack of risk-based AML software tools, since at that time, market for AML software offered rule-based systems. (Le Khac *et al.*, 2016). This was, of course, before the emergence of artificial intelligence and the unleashing of numerous AI risk-based AML solutions. In addition, before the EU's Fourth AML Directive was drafted, the rationale behind it was the need to strengthen the existing EU anti-money laundering system, due to the technological advancements that have rapidly shifted the landscape in which criminals operate, but also in which they can easily adapt (Cotoc *et al.*, 2021).

5.2. Relevant definitions

Before delving into practical applications and the potential of utilizing AI for AML purposes, key terms need to be presented. First of all, although numerous definitions of AI exist, this paper adopts the definition provided by the EU AI Act, which describes AI as machine-based systems capable of inferring information through learning, reasoning, or modeling, going beyond basic data processing. In addition to this definition, key terms that need explaining are machine learning, natural language processing, agentic and generative AI, but without going outside the regulatory scope of the paper and without delving into technicalities.

Therefore, machine learning is a subfield of artificial intelligence that gives computers the ability to learn without explicitly being programmed (MIT Sloan, 2021). Natural language processing (NLP) is meanwhile described as another subset of AI, enabling machines to analyze and interpret human language in text form, with automation of repetitive tasks and improved data analysis and insights being underlined as the main benefits of NLP (IBM, 2025).

Other key terms include generative and agentic AI, which are two distinctive forms of artificial intelligence. Generative AI creates content based on specific data and instructions (prompts), while Agentic AI is autonomous in the sense that it is based on goal-oriented behavior and tools to execute multistep tasks and make decisions (Thomson Reuters, 2025). Both of these forms show high potential in handling money laundering cases, specifically, generative AI can

be used to track financial-crime patterns, but the dataset that the generative AI is "fed" with is crucial. In addition to this, Agentic AI can perform a higher span of tasks due to its autonomous decision-making features, which can especially be useful for autonomous transaction monitoring and analytical overviews, which could again be useful for compiling STRs. However, with the Agentic AI, a crucial component is human oversight, as autonomous decision-making needs to be controlled with human-based experience in compliance. This is especially important in order to avoid any mistakes that could lead to reputational risks and a decrease in public trust.

5.3. Possibilities of utilizing AI for AML

Despite the numerous possible applications of AI in AML, and based on the needs of the AML compliance industry, the main usages of AI in AML include enhancing customer due diligence (CDD) and improving transaction monitoring, as these processes rely mostly on data entry and its monitoring. In addition, due to the existing "legacy systems" (FATF), data entry is mostly manual, which makes the work of compliance officers and investigators cumbersome. All of the mentioned can be fully alleviated with AI tools.

Furthermore, AI can enable more effective risk assessments through a risk-based approach (rather than a rule-based approach). Once again, FATF underlined that these tools can serve as an alternative to the traditional risk analysis, which is carried out by compliance officers. Therefore, AI should help, not replace seasoned compliance officers. On that note, it is important to underline the openness of seasoned AML compliance officers and experts to the usage of AI, as they deem it an inevitable solution to current AML system deficiencies, but they also put a strong emphasis on the necessity of empirical value that experienced AML officers have in assessing money laundering threats.

More to this point, some authors stated that the ratio between data assembling and actual analytical reasoning needs to be shifted, so that more time is spent on utilizing these traits rather than spending time on repetitive tasks such as manual data entry (Richards, 2020). This also means, from a philosophical point of view, a more dignified labor in terms of real value one brings to the compliance world, rather than mere data entry, which is one of

the key areas that can be automated via AI tools. Hence, instead of the current ratio, where most of an AML analyst's time is spent on assembling the data, with little time left for analytical thinking based on their experience and on uniquely human traits, such as empathy and instinct, the key is to flip this ratio. In that manner, analysts can spend less of their time on handling data and most of their time thinking and analyzing, providing value that a machine simply cannot. (Richards, 2020)

5.4. International perspective on using AI for AML

Financial Action Task Force (FATF) Global Network and Egmont Group of Financial Intelligence Units (Egmont Group) have also recognized several needs of the current AML/CFT system that can be met with AI tools. For instance, they've entailed that there is a need to make the manual process of data gathering from a large number of entities in a more streamlined manner, which would as a result, also facilitate a more efficient processing of data by FIUs. In addition to this, allocation of resources is another key aspect when advocating for the application of AI in AML efforts. Precisely, with the automation of data sorting, human resources would be relieved of these mundane tasks and focused on more analytical tasks. Furthermore, FATF and EGMONT entail that suspicious transaction reports (STRs) can benefit most from natural language processes, since patterns can be identified (and even new forms of money laundering), which can be helpful for FIUs to prepare high-quality analytical reports. Finally, it is emphasized that the potential of AI tools also lies in the identification of new risks which were previously unknown (FATF; Egmont, 2021).

Besides, AI-powered AML systems should be considered by financial investigation units, as FIUs are incumbent to domestic coordination, and domestic coordination is crucial for proper international cooperation in combating money laundering (FATF, 2025). Even in those jurisdictions that do not have in place proper technological tools, but with effective and efficient domestic coordination, authorities could benefit from the tools available in other jurisdictions, via mutual (international) legal assistance or joint investigative teams on a specific money laundering case. Therefore, technological advancement is not an unachievable goal – it is very real to consider its utilization even by those countries that did not meet certain

standards in the evaluation rounds as defined by FATF. In fact, by utilizing these tools that exist in other, more advanced countries when it comes to money laundering prevention, through mutual legal assistance and international cooperation channels, there could be drastic improvements domestically.

6. Case studies

6.1. Vendors' AI-AML solutions

There are numerous examples of AI-powered AML solutions that were deployed either by obliged entities (offered by a specific vendor of the AI tool) or developed through international efforts to combat money laundering. In addition to this, certain pilot exercises also proved the potential that the application of AI has in enhancing AML efforts.

There is an almost overwhelming number of AI AML tools that are offered via different vendors, which is attested by the fact that the market value of vendor offerings for AML software is estimated to reach \$5.91 billion by 2032. This resulted in obliged entities, such as banks, compliance firms, or the private sector, deploying tools that will ease the everyday compliance processes.

For instance, SEON has adopted an AI-driven transaction monitoring system, leading to notable gains in detection accuracy and operational efficiency. real-time risk and AML decision-making, helping businesses detect hidden threats and automate compliance. Just recently, it was published that SEON introduced novel features that tend to ease the compliance burdens, such as the Movement of Funds tool, which visualises transaction flows to identify layering schemes and structuring patterns (RegTech Analyst, 2023). This kind of advanced technology can help mitigate not only well-known risks, but also recognize new risks.

6.2. International projects on utilizing AI for AML

As for the international efforts in deploying such tools, there were also various successful initiatives. In Peru, the Financial Intelligence Unit, with GIZ, launched Inspector AI, which analyzes structured and unstructured data in suspicious transaction reports, ranks cases by risk, and has led to more than

double the number of cases referred for prosecution. of cases that could be referred to prosecutorial authorities (Zinnbauer, 2025). This initiative is a successful example of deploying tools that will ease the burden of data entry for compliance officers, allowing them to focus on priority (high-risk) cases (GIZ, 2024).

On the EU region scale, the Datacorss tool developed by the Transcrime research centre has been adopted by public authorities in several countries (e.g., Romania, France, Lithuania) to conduct risk assessments and detect anomalies in firms' ownership structures that can flag risks of collusion, corruption, and money laundering. As the latest in the series of Datacross actions, Datacross III tends to develop and implement advanced AI and machine learning models to improve the detection of high-risk entities and transactions, while minimizing false positives, as well as the detection of illicit financial activities with the aid of AI-driven anomaly indicators (Datacross III, 2024).

6.3. Project Aurora

Finally, another instance that showcases the high potential of applying artificial intelligence to anti-money laundering is the Project Aurora. This project concerns not the actual deployment of an AI-powered AML compliance system, but rather a proof-of-concept that explored how the detection of illicit financial flows doesn't need to be carried out in a "siloed" manner, by separate financial institutions, but through collaboration of multiple financial institutions, all to detect hidden and cross-border money laundering networks.

The Project team used synthetic data on payments, both domestic and international ones, with embedded money-laundering elements and applied several machine-learning models to different monitoring scenarios - at a siloed financial institution level, a national level, and a cross-border level. The results of the simulation were impressive. Foremost, the exercise showed that machine learning in AML compliance outperforms traditional rule-based monitoring by far, with ML models detecting more money-launderers than the rule-based systems. In addition to this, a common mishap in monitoring money laundering data is the false positives, which were shown to be reduced by 40 % when institutions monitored data individually and by about 75 % when

they pooled data at the national level. The Project also focused on common privacy risks when it comes to applying AI to AML, advocating for the use of privacy-enhancing technologies to protect sensitive data while enabling analysis. In addition to this, Project Aurora aims for advanced AI and data sharing between financial institutions in order to improve global efforts in combating money laundering, but with prior safeguards such as legal frameworks on privacy and access to high-quality data (Bank for International Settlements, 2023).

7. Risks specific to AI utilization in AML

7.1. Risks specific to the AML regime

In every conversation surrounding the potential of using AI in any field, there is an undeniable factor that rightfully minimizes the AI enthusiasm. This factor is the inherent risks that are associated with its utilization. In a field such as money laundering, where the sheer amount of personal (financial) information that forms the basis for the AI model dataset, these risks are not to be taken lightly. Even though the AML field welcomes the use of AI to increase its overall effectiveness, the AML system already has its own pitfalls that could lead to reputational risks, a consequence aimed to be avoided in each stage of the compliance process.

For instance, traditional anti-money laundering systems produce a high rate of false positives (the so called “infamous 95%), meaning most alerts they generate do not correspond to real threats. In other words, these systems frequently flag legitimate transactions as suspicious (PWC, 2010; KPMG, 2022). Therefore, AI should build exactly on these weak areas of the current AML system to mitigate them and show that technological advancements are the way to go forward, and not a liability that will create additional weak spots.

7.2. Risks specific to AI

In addition to this, as with AI tools in any field, there are some essential risks specific to how artificial intelligence and machine learning operate. As mentioned in the beginning, datasets are the basis for the deployment of AI tools. Therefore, improper and biased data will most certainly lead to hindrance in obtaining proper results. A notable example occurred in 2023

when an AI system deployed by a multinational bank flagged 60% of transactions from a particular region as high-risk. Upon investigation, it was found that the system was trained on biased data that disproportionately associated specific geographies with financial crime (Lucinity, 2025).

Another risk, which is linked not only to AI tools developed for compliance purposes, but any AI tool, is the inherent fault all AI systems have - the replication of the developers' own biases. As pointed out in the FATF report on opportunities of emerging technologies for AML, this would mean that a developer could (consciously or unconsciously) tag certain financial activities as suspicious based on the biased risk-profiling of certain types of individuals or entities, disabling their access to certain financial products and services. This can be mitigated through developing responsible and ethical AI systems, minimizing risks such as bias, discrimination, a lack of transparency, and breaches of data privacy. (FATF, 2021). If, for instance, an AI system used for AML purposes sources data from certain obliged entities that have applied de-risking rather than applying comprehensive and continuous risk assessment, this, in turn, could also mean that the same system would de-risk in the same manner. That is why the key takeaway in including AI systems for AML is data sourcing.

Testing AI-powered AML systems would be ideal for starters, as obliged entities could engage compliance officers to train the model and "feed" the AI with a proper set of data. Then, continuous oversight of the functionality of the system would need to be set in place, which could be introduced as part of the broader AML compliance policies by obliged entities.

In addition, as showcased in the case studies part, the vendors' market of technical solutions to AML is growing rapidly. Therefore, there is a legitimate concern that this trend has the potential for possible misuse by vendors. This concern was also posed by FATF, in which they stated that vendors would need to go through additional controls in order to minimize possible risk. The main question that FATF poses is whether providers (vendors) of these tools should be separately scrutinised.

7.3. Risk-mitigation regulatory framework

Relevant standards on regulating AI, such as the EU AI Act, entail that AI systems specifically intended to be used by financial intelligence units carrying out administrative tasks, analysing information for AML, should not be classified as high-risk AI systems. Therefore, it stems from the wording of the EU AI Act that AI systems used for AML do not infringe on fundamental rights (as this is a prerequisite for an AI system to be classified as high-risk per the EU AI Act). Still, the same Act entails that suspects (during the investigation phase) and later on defendants before the court have the right to obtain information on how the system operated as well as to challenge the results of it. This goes directly along the lines of transparency and explainability principles.

Finally, the above-mentioned risks can be mitigated through a set of robust principles that aim to ensure trustworthy AML systems. This can be achieved with human oversight, as it is crucial to a successful deployment and control of any AI system, including the ones utilized for AML. In the context of AML, the human oversight would ideally be conducted by experienced compliance officers, making sure that any anomaly in the work of the AI system is detected and reported, in order to make the AI learn from proper data.

As for the mentioned FATF concern on regulation of vendors, even though this is not regulated for the AML system specifically, the EU Regulation 2024/1773, which supplements the Digital Operational Resilience Act – DORA, includes requirements for the contractual clauses on mutual obligations of the financial entities and the ICT third-party service providers, which should be set out in writing.

Other relevant international standards in the field of AI governance, such as the OECD AI Principles, promote the use of AI that is innovative and trustworthy, rooted in transparency and explainability. As for these two key principles, one of the crucial safeguards in using AI in AML is avoiding the so-called "black-box", and enabling compliance officers/investigators to understand why a specific transaction or a customer was flagged as suspicious (Ires, 2024). This lack of transparency can hinder the ability of compliance

officers to investigate and escalate cases effectively. In addition, the UNESCO Recommendation on the Ethics of Artificial Intelligence calls for principles such as accountability and human oversight to ensure that AI remains within the control of humans. For AI utilized for AML purposes, human oversight is crucial.

8. Conclusion

This paper presented the potential of artificial intelligence tools in anti-money laundering, through the lens of the ineffectiveness of the current anti-money laundering regime. The main causes of this ineffectiveness are rooted in high compliance costs for obliged entities, which lead to cumbersome administrative burdens as well as compliance costs for clients, leading to delayed transactions due to regular client identification processes. In addition, a high number of transactions result in false positives, meaning they are flagged by the system but are a dead end when it comes to actual money laundering cases. This results in numerous such transactions being reported, with the financial intelligence units not being able to compile suspicious transaction reports that could lead to successful money laundering investigations. In addition to this, due to the nature of money laundering and its ability to adapt and change forms, which is something highly exploited by criminals (as reported by EBA), especially in the light of new technological developments, it makes it harder to trace. That is why the main course of action in this changed ML environment is countering it with the same tool – artificial intelligence.

As presented in the paper, AI holds great potential in improving the overall response of the AML system. Without advocating for any type of solution, either generative AI or agentic AI-based, the paper merely entails the benefits of each, while specifying that agentic AI, due to its autonomous features, could ease the daily work of compliance officers/investigators. This could result in what many authors point to as the basic rationale behind adopting AI in AML – releasing human resources from mundane tasks, and redirecting those resources to more complex and analytically intricate cases. This leads to the point of human oversight as a crucial safeguard. With the human-in-the-loop safeguard, AI-based AML systems will ensure that the risks associated with

either AML or AI are mitigated. Hence, AI should serve as a complementary enabler of AML efforts, not intended to replace strong institutions, skilled investigators, and compliance officers, as well as robust legal frameworks.

Finally, in order to maximize the benefits of using AI in AML and minimize risks, a cost-benefit assessment of using AI in AML is crucial, as this is something that usually curbs enthusiasm for its utilization, as entailed by FATF. Also, regardless of the existing general rules on AI governance, a comprehensive regulatory framework for AI in AML should be adopted for the same cause as doing the cost-benefit analyses - minimizing risks while maximizing benefits of using AI for AML purposes. As presented in this paper's case studies, the potential of AI in alleviating AML workload is enormous; however, without robust guidelines, there could be more damage done. In the end, comprehensive guidelines in the so-called RegTech (regulatory technology) and SupTech (supervisory technology) are the key in ensuring that technologies don't bring about more havoc than benefits in the already fragmented AML system.

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